

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

1. Connector & input protection

Field connector, 2-stage bidirectional TVS surge protection, chassis isolation

File: HMT500(260313A)_connector.kicad_sch

2. Power

eFuse (reverse/OV/UV), 60V buck to 5V, LDO 3.3V

File: HMT500(260313A)_power.kicad_sch

3. MCU

STM32G0B1CCT3, SWD, reset, internal temperature sensor

File: HMT500(260313A)_mcu.kicad_sch

4. Measurement

Capacitive humidity sensor (PCAP04) and Pt1000 2-wire (ADS1220)

File: HMT500(260313A)_measurement.kicad_sch

5. Analog outputs

2x DAC8760 V/I output with TPS26611 miswiring protection

File: HMT500(260313A)_analog_out.kicad_sch

6. RS-485

THVD2450 +/-70V fault-protected transceiver

File: HMT500(260313A)_rs485.kicad_sch

DOTECH HMT500(260313) – Oil Moisture Transmitter – Schematic v0.8 – Project No. HMT500(260313A)

File names (schematic, PCB, Gerber) = HMT500(260313A).* PCB silkscreen marking = HMT500(260313A)

v0.5: J1 = JST GH 1.25 mm 8P side entry (SM08B–GHS–TB), harness W–2 to M12 8P; J5 chassis spring contact (was M12 shell pin).

v0.4: J3 = JST SH 1.0 mm 4P (SM04B–SRSS–TB side entry, v0.5) for sensor harness W–1 direct from HTX99R (no feedthrough, mech Rev F); Pt1000 2-wire (Kelvin split at J3 pads).

v0.3: MCU STM32G0B1CCT3 (LQFP48, –40..125 C, in-house part); TPS26600PWP official pinout; single axial PCB 57x23 (mech Rev C).

Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2450 -> J1

Outputs: RS–485 Modbus RTU + 2x analog (4–20 mA / 0–20 mA / 0–10 V / 0–5 V selectable). Supply 12–28 V DC.

Protection target: any pin pair +/-30 V continuous (miswiring), surge +/-1 kV, ESD +/-8 kV contact.

Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), CHASSIS. Inter-sheet signals: global labels.

v0.6: parts cleanup (–16: pull-ups R11–15, LED D3/R16, RS–485 R60/61 D60/61, FB2 C19 C11 C2 C6); PCB temperature = ADS1220 + MCU internal sensors.

v0.7: DAC8760 datasheet pinout (HTSSOP–24 PWP); +VSENSE via OPA197 follower (U12/U13); external ISET–R 15k 0.1%; DACs on SPI2 daisy chain, common DAC_LATCH.

v0.8: datasheet review (docs/hw/schematic–review–260313A.md): real pinouts LMR36006/PCAP04/TPS26611; TPS2660 RTN isolated from GND; OVP 29 V (supply 12–28 V);

DAC SPI = gated SCLK (daisy chain removed in DAC8760 rev D); DAC AVDD 10R; ALARM wired–OR + 10k; TVS3301 outputs; PCAP04 on SPI3; buck L/C per datasheet.

TBD footprints: J5 chassis contact, CMC, GDT, buck inductor L2, LMR36006 (RNX0012A 2x3 mm, not in KiCad lib).

표기 규칙: 부품번호 = 굵은 글자 (R1, U4) / 부품값 = 보통 글자 (10k, DAC8760) / 전원 네트 = 기울임 글자 (+3V3, VIN_P)

신호 네트 = 테두리 있는 라벨 (SPI_SCK, OUT1_EXT) - 네트 이름은 부품번호·부품명과 겹치지 않게 지음

글자 편집: 생성기가 부품 몸체 핀·선·라벨·메모와 겹치지 않는 자리에 자동 배치하고, 겹치면 생성 실패로 처리

Pinouts checked vs manufacturer datasheets 2026–09–30 (STM32: KiCad lib)

Design notes: docs/hw/circuit–design.md

Project No. HMT500(260313A) – HMT500(260313) oil moisture transmitter

DOTECH Co., Ltd.

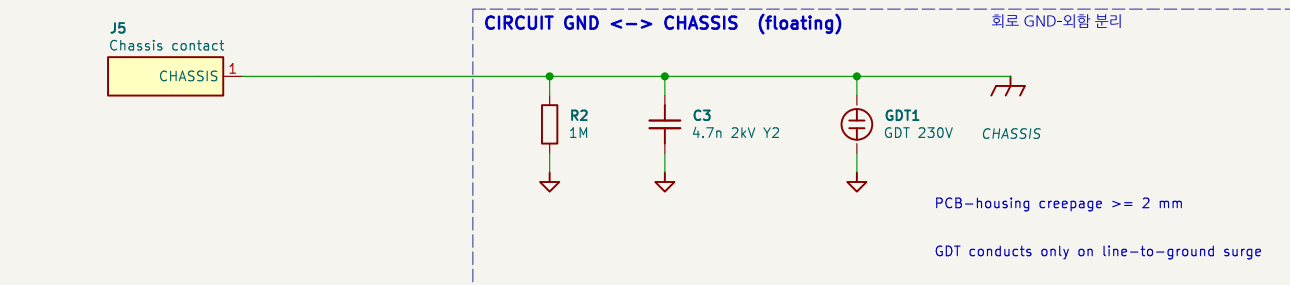
Sheet: /

File: HMT500(260313A).kicad_sch

Title: HMT500(260313A) Top

Size: A3Date: 2026–09–30Rev: 0.8

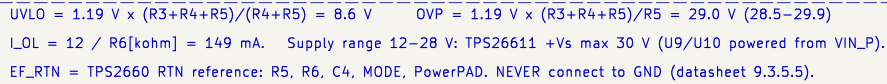
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4Id: 1/7



Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
 Design notes: docs/hw/circuit-design.md
 Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter
DOTECH Co., Ltd.
 Sheet: /1. Connector & input protection/
 File: HMT500(260313A)_connector.kicad_sch

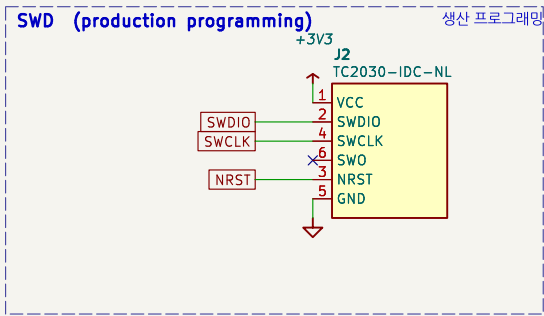
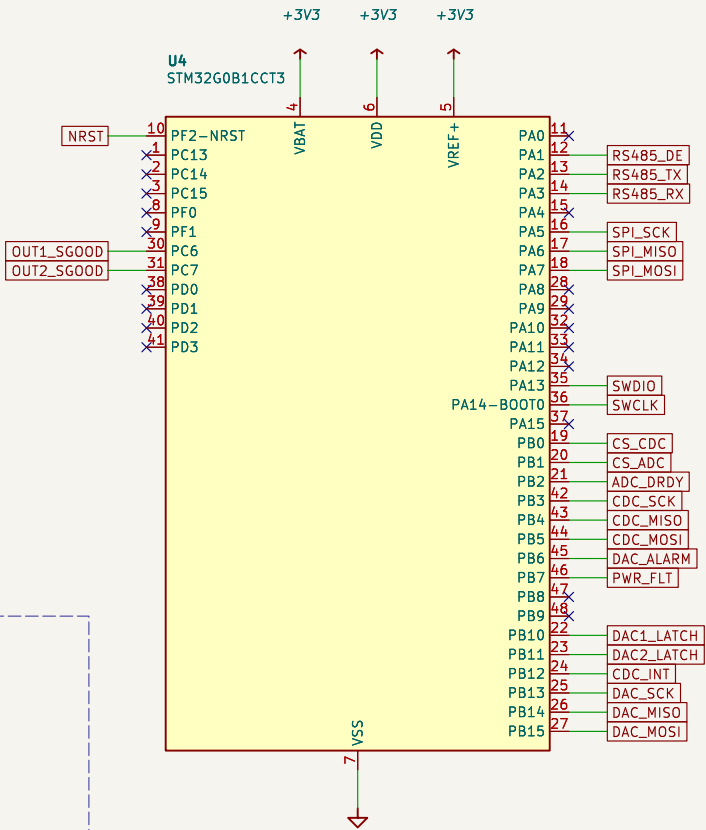
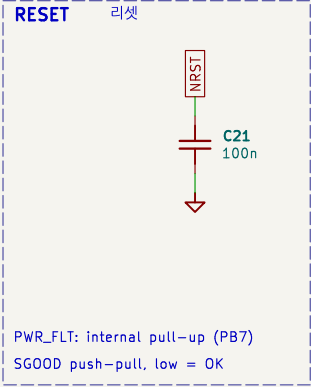
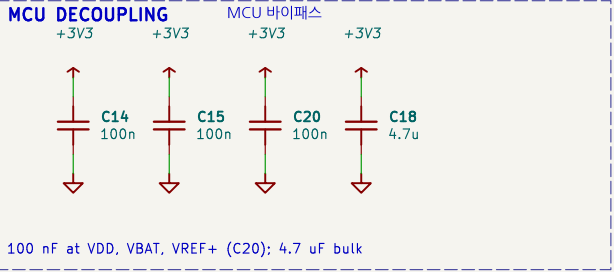
Title: HMT500(260313A) Connector & input protection

Size: A3	Date: 2026-09-30	Rev: 0.8
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/7



* 표시 값은 데이터시트로 확인

	7	8
--	---	---



MCU NOTES (STM32G0B1) MCU 설정 메모

BOOT0 shares PA14/SWCLK - no pull-down.

Option bytes: nBOOT_SEL=1, nBOOT0=1

-> boot from flash (RS-485 bootloader 0x08000000).

VDDA = VDD (LQFP48); VREF+ tied to +3V3 (C20).

PCB temp (mid): internal sensor, ADC ch TS + TS_CAL.

Cortex-M0+: no SWO (J2 pin 6 NC).

Same die as DP2000 (G0B1CCT6); T3 = -40..125 C.

주요 부품 — 역할·특징·사양

부품번호	부품	역할	특징·사양
U4	STM32G0B1CCT3	제어·통신·보정 연산	Cortex-M0+ 64 MHz, 플래시 256 KB, RAM 144 KB, -40-125 °C, LQFP48, 내부 온도센서 (PCB 온도). SPI1 = ADS1220, SPI2 = DAC (모드 3, SCLK 게이트), SPI3 = PCAP04
J2	TC2030-IDC-NL	생산 때 펌웨어 기록 (SWD)	PCB 패드만 (부품 없음). 이후 업데이트는 RS-485 부트로더
C21	100n	리셋 노이즈 필터	NRST 핀, 물딩 후 EMC 여유

* 표시 값은 데이터시트로 확인

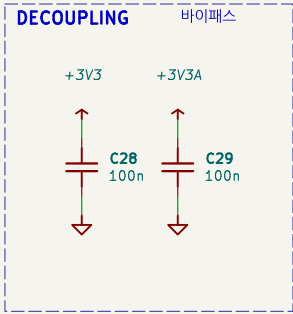
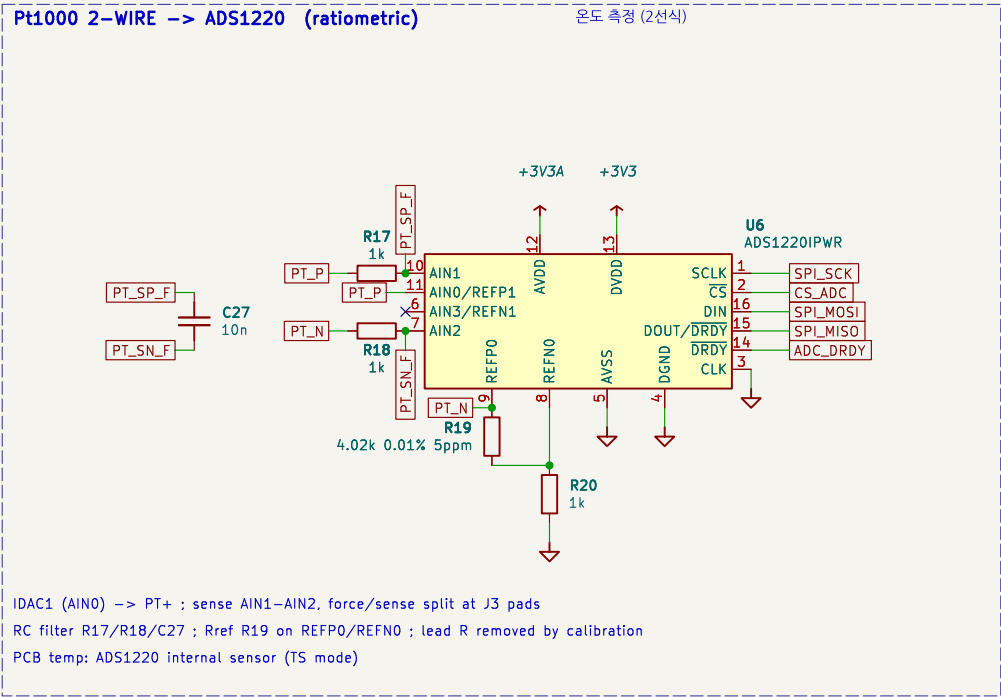
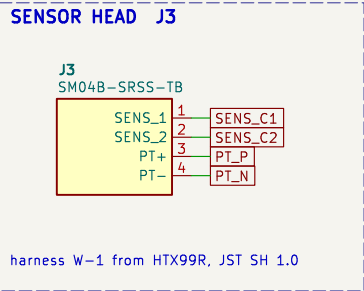
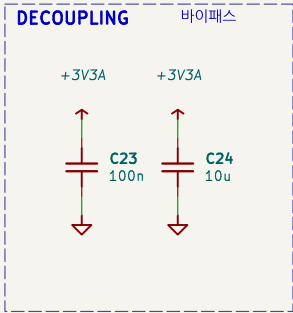
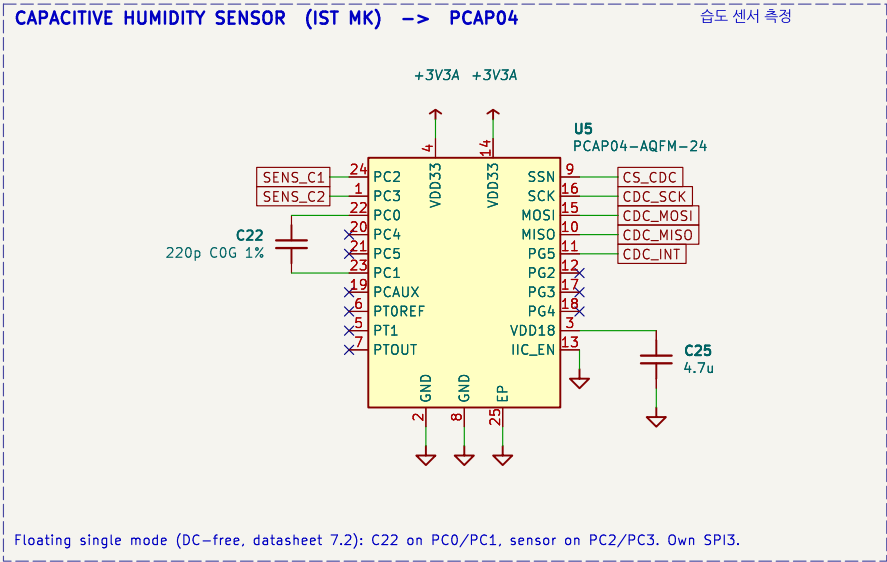
Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter

DOTECH Co., Ltd.

Sheet: /3. MCU/
File: HMT500(260313A)_mcu.kicad_sch

Title: HMT500(260313A) MCU

Size: A3	Date: 2026-09-30	Rev: 0.8
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7



주요 부품 — 역할 · 특징 · 사양			
부품번호	부품	역할	특징 · 사양
J3	SM04B-SR5S-TB	센서 하네스 W-1 연결	JST SH 1.0 mm 4P 옆 삽입, 1-2 = MK33, 3-4 = Pt1000
U5	PCAP04-AQFM-24	습도 센서 정전용량 측정	정전용량-디지털 변환, 기준 C와 비율 측정(방전 시간), DC 없는 구동, 내부 DSP, 전용 SPI3 (모드 1)
C25, C24	4.7u / 10u	PCAP04 전원 버퍼	VDD18 ≥ 4.7 uF, VDD33 ≥ 10 uF (데이터시트 표 2)
C22	220p COG 1%	PCAP04 기준 용량	MK33-W mini (200 pF)와 비슷한 값, COG (온도 변화 거의 없음)
U6	ADS1220IPWR	Pt1000 온도 측정	24비트 ADC, 50/60 Hz 제거, 내부 온도센서 (PCB 온도), 설정 한계: IDAC ≤ 250 uA, PGA ≤ 2 (SPI1 모드 1)
R19	4.02k 0.01% 5ppm	Pt1000 비율 측정 기준저항	IDAC 전류 오차 상쇄, 온도계수 5 ppm/°C
R17, R18, C27	1k / 10n	입력 RC 필터	차동-공통모드 노이즈 제거
* 표시 값은 데이터시트로 확인			

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter
DOTECH Co., Ltd.

Sheet: /4. Measurement/
File: HMT500(260313A)_measurement.kicad_sch

Title: HMT500(260313A) Measurement

Size: A3	Date: 2026-09-30	Rev: 0.8
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/7

